

- Uses flexible and intuitive APIs in a Web browser to build models from scratch with the aid of the low-level JavaScript linear algebra library or the high-level layers API.
- Develops ML in Node.js
 - Using Node.js runtime, we can execute native TensorFlow with the same TensorFlow.js API
- Runs existing models
 - With TensorFlow.js model converters, we can run pre-existing TensorFlow models right in the browser
- Retrains existing models
 - Allows retraining of the pre-existing ML models using sensor data connected to the browser or other client-side data

With over 13,000 stars and almost 1,100 forks at the time of writing this article, it is also one of the most popular and actively maintained ML frameworks on GitHub.

Note: The TensorFlow.js framework is open source and the source code of the framework is available on GitHub. Developers/programmers are encouraged to participate and contribute to the project.

face-api.js

face-api.js is a JavaScript module that implements convolutional neural networking to solutions in the face detection and recognition space as well as for facial landmarks. face-api.js leverages TensorFlow.js and is optimised for the desktop and mobile Web. With over 9,400 stars and 1,700 forks, it is also one of the popular face detection and face recognition open source JavaScript APIs on GitHub.

There are several pre-trained models available with face-api.js, including face detection, facial landmark detection, face recognition, facial expression recognition, age estimation and gender recognition. We

will use some of them in our app.

Installing the prerequisites

Before we start developing the app using TensorFlow.js or face-api.js, we need to install Node.js as the TensorFlow.js and the face-api.js framework, which we are going to use to detect a face and its expressions. It is also distributed as an *npm* package, and this makes it easy to set up the project's structures to develop ML apps.

Note: It is assumed that you have some basic knowledge of Web technologies like HTML, CSS and JavaScript. If you don't, W3Schools (<http://www.w3schools.com/>) is a good place to start. The site has some great tutorials for Web technologies, and they are easy to follow.

Installing Node.js

Download and install Node.js (<https://nodejs.org/>). TensorFlow.js and face-api.js are also distributed as *npm* packages, so we will use Node and *npm* to install the TensorFlow.js and face-api.js frameworks.

Creating the app folder

Create a folder for your project, and change the directory to it. Run the following commands at the prompt or in the terminal:

```
mkdir face-detect
cd face-detect
```

Initialise the folder using the following *npm* command:

```
npm init -y
```

This will create a *package.json* manifest file with important default information about the project, and will add the necessary node modules.

Installing face-api.js

We can install face-api.js as a node package in the project folder. Just run the following command in the terminal or at the command prompt:

```
npm install face-api.js --save-dev
```

Installing HTTP server

To run the project, we need to install an HTTP server. Run the following command in a terminal or at the command prompt:

```
npm install http-server --save-dev
```

Setting up the folder structure for the project

Create the following files and folders, some of which will have been automatically created by the commands run above.

- *index.html* – In this folder, we will create the UI of the app.
- *script.js* – In this folder, we will add the logic of the app.
- *models* – In this folder, we will add pre-trained models for

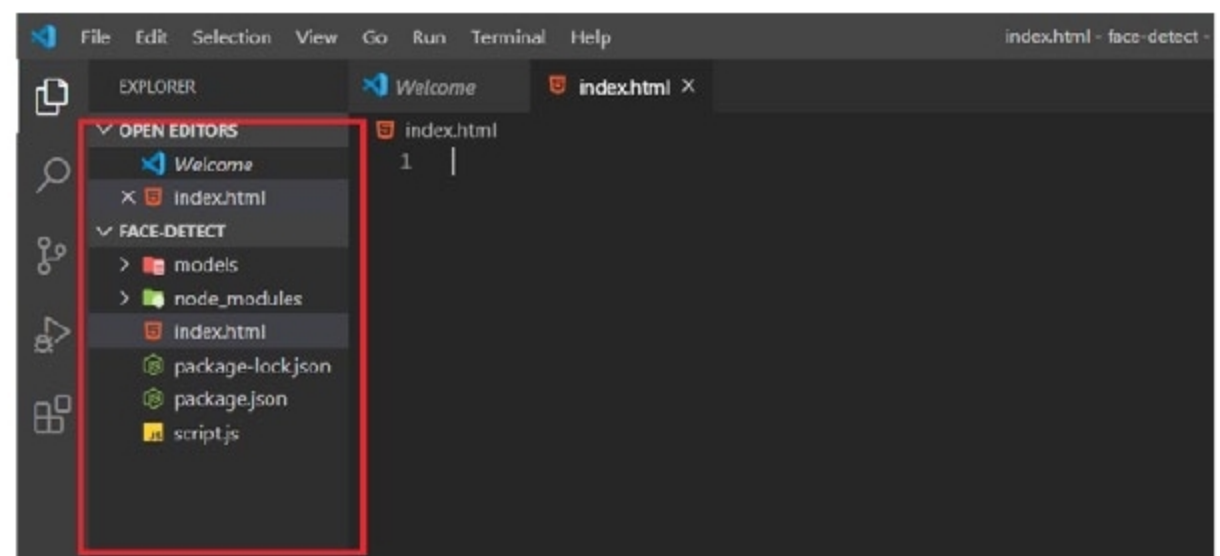


Figure 1: Project file structure